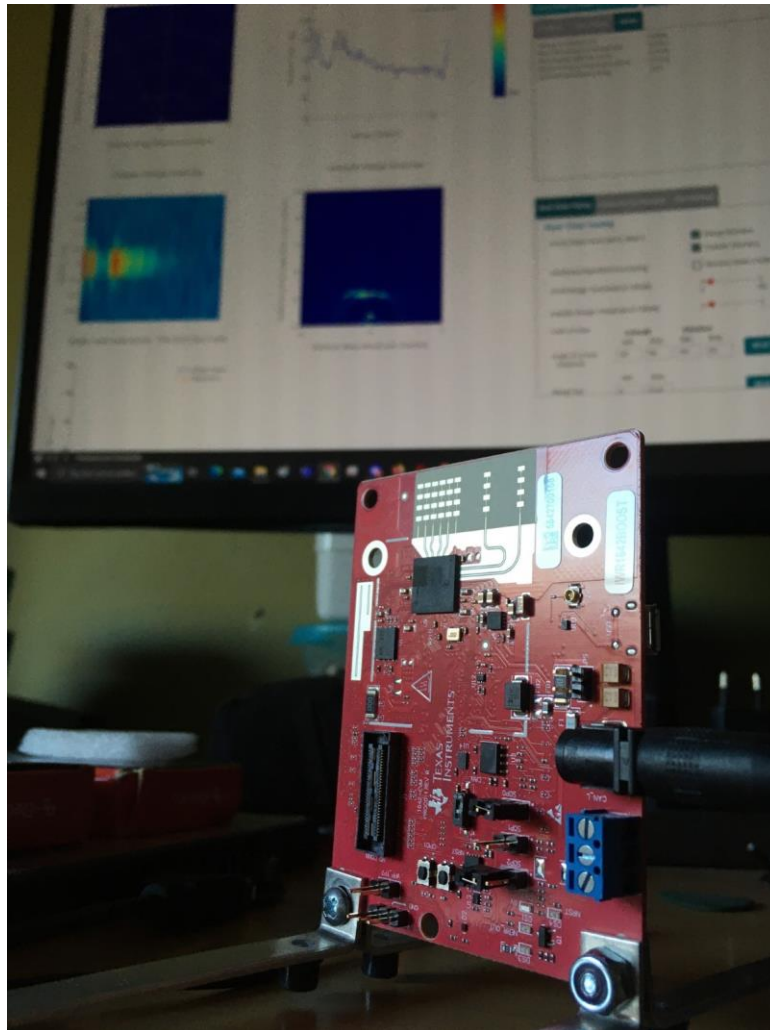


TI IWR1642boost Range And Velocity



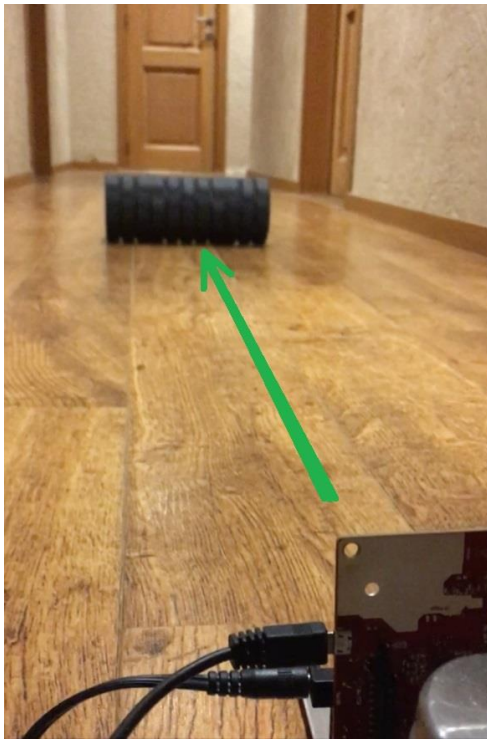
Setup material:

- TI IWR1642boost module
- TI IWR1642 Out-Of-The-Box Object Detection Demo software
- USB connected laptop running the TI Demo Visualizer in Chrome
- Target object: Rolling foam roller

Goal:

- Test the complete TI IWR1642 toolchain
- Get the very first measurement data out of the device
- Play around with different configurations using the mmWave Demo visualizer to gain a first hands-on experience with the TI IWR1642 basic functionality

TI IWR1642 Measuring Setup:



In this setup, a fitness foam roller is used as the target object that will move away from the TI IWR1642boost module under an angle of $\pm 0^\circ$. The target is placed a couple of centimeters away from the TI IWR1642boost module as a starting point. A slight push will make the foam roller roll down the hallway in a straight line.

TI IWR1642 Measuring Results – mmWave Demo Visualizer:

Main Configuration of the TI IWR1642

Note: These configuration values are chosen on trial and error because the default settings did not work well with the used test setup. Future examples will use actual calculations for optimal settings.

mmWave Demo Visualizer | Options | Help

Configure

Setup Details

Platform	xWR16xx
SDK version (*)	3.6
Antenna Config (Azimuth Res - deg)	4Rx,2Tx(15 deg)

RCS

Desired Radar Cross Section (sq. m)	0.2
Maximum Range for desired RCS (m)	36.011
RCS at Max Unambiguous Range (sq. m)	0.000819

Desirable Configuration

Best Velocity Resolution	
Frequency Band (GHz)	77-81
Calibration Data Save/Restore	None 0x1F0000

Scene Selection

Frame Rate (fps)	10
Range Resolution (m)	0.178
Maximum Unambiguous Range (m)	9.11
Maximum Radial Velocity (m/s)	2.56
Radial Velocity Resolution (m/s)	0.04

Plot Selection

☒ Scatter Plot	☐ Range Azimuth Heat Map
☒ Range Profile	☐ Range Doppler Heat Map
☐ Noise Profile	☒ Statistics

SEND CONFIG TO MMWAVE DEVICE

SAVE CONFIG TO PC

RESET SELECTION

Console Messages

```
Done
mmwDemo:/>measureRangeBiasAndRxChanPhase 0 1.5 0.2
Done
mmwDemo:/>CQRxSatMonitor 0 3 4 31 0
Done
mmwDemo:/>CQSigImgMonitor 0 31 4
Done
mmwDemo:/>analogMonitor 0 0
Done
mmwDemo:/>aoaFovCfg -1 -90 90 -90 90
Done
mmwDemo:/>cfarFovCfg -1 0 9.11
Done
mmwDemo:/>cfarFovCfg -1 1 -2.49 2.49
Done
mmwDemo:/>calibData 0 0 0
Done
mmwDemo:/>sensorStart
Done
mmwDemo:/>cfarCfg -1 1 0 8 4 4 1 5 1
Done
mmwDemo:/>aoaFovCfg -1 -45 45 -90 90
Done
mmwDemo:/>cfarFovCfg -1 0 0 8
Done
```

CLEAR CONSOLE

Real-Time Tuning | Advanced Commands | Plot Settings

Real-Time Tuning

Additional Algorithm Processing	☐ Remove Static Clutter
CFAR Range Threshold (0-100dB)	100
Doppler Range Threshold (0-100dB)	100
Field of View	
Angle of arrival (degrees)	Min: -45, Max: 45, Min: -90, Max: 90
Range (m)	Min: 0, Max: 8
Doppler (m/s)	Min: -2.49, Max: 2.49

RESET

SEND

RESET

SEND

RESET

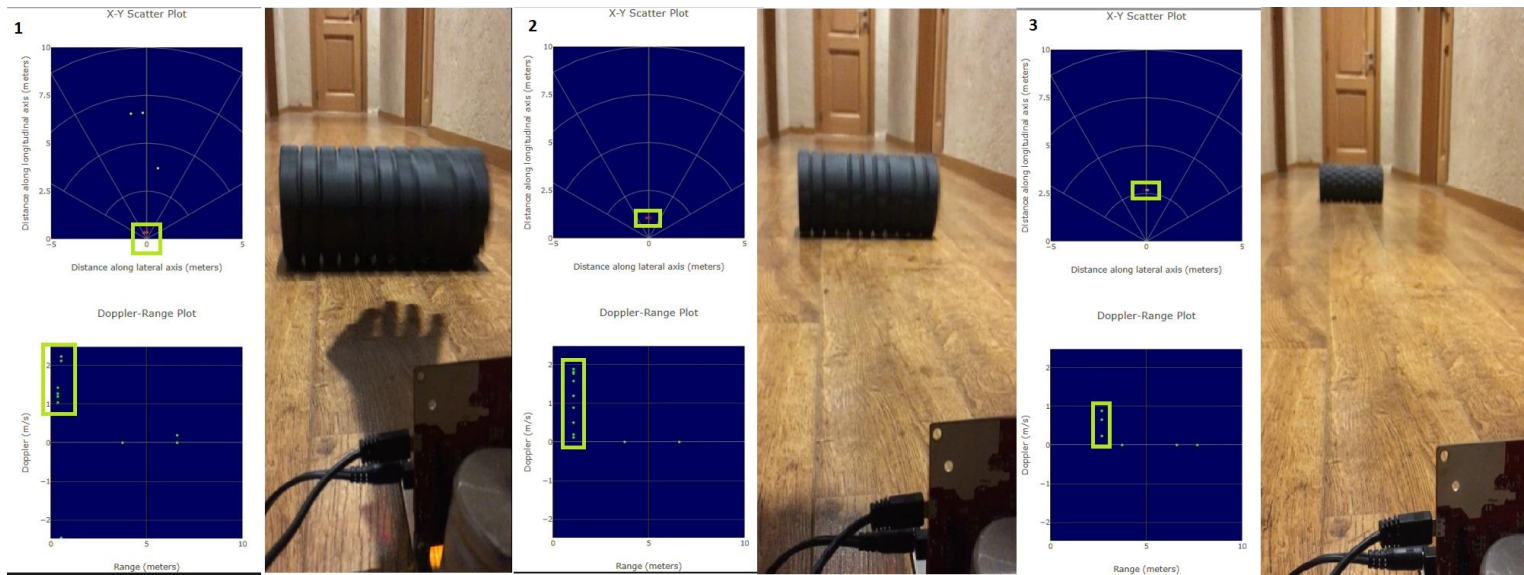
SEND

Change Plot Settings

Scatter Plot Display Time (0-0.25secs)	0.10
Scatter Plot Colormap	Velocity

Processed Measurements:

See TI_IWR1642_RollingFoamRoller.mp4 in the ObjectDistanceAndVelocity folder for a video demonstration.



Results:

- The rolling action of the foam roller causes a notable change in the X-Y scatter plot. A group of points are increasing in distance while staying around an angle of 0° with respect to the TI IWR1642boost module.
- The Doppler Range plot shows a notable group of points where movement/velocity is detected. These velocity points also increase in range with respect to the TI IWR1642boost module.

Conclusion:

The moving object is clearly notable in this test. Even when no calibration of the TI IWR1642 and no measurement compensation is done, this test still shows a decent example of the TI IWR1642 functionality.

As the main goal of this demo project is to test the TI IWR1642 toolchain and demonstrate some of the functionality of the IWR1642 and not the accuracy of measurements, it can be concluded that this Demo project succeed in its intention.